

f -Divergence:

Given 2 probability Distribution func with the corresponding density func
(P, Q) (p, q)

then the f -Divergence b/w them is defined as

$$D_f(P||Q) = \int_X q(x) f\left(\frac{p(x)}{q(x)}\right) dx.$$

$f(u) : \mathbb{R}^+ \rightarrow \mathbb{R}$ Convex, Left Semi Continuous, $f(u) = 0$

$$D_f(\cdot || \cdot) \geq 0$$

$D_f(\cdot || \cdot) = 0$ only when they are identical

$$f(1) = 0$$

Assume P is absolutely continuous wrt Q so that $p(x)/q(x)$ is well defined whenever needed.

$$\text{Likelihood ratio } R(x) = \frac{p(x)}{q(x)}$$

f is convex & we can use Jensen's inequality

$$\mathbb{E}_Q [f(R)] \geq f(\mathbb{E}_Q [R])$$

$$\mathbb{E}_Q [R] = \int q(x) \frac{p(x)}{q(x)} dx$$

$$= \int p(x) dx$$

$$= 1$$

$$D_f(P \parallel Q) = \mathbb{E}_Q[f(R)]$$

$$\geq f(\mathbb{E}_Q[R])$$

$$= f(1)$$

$$= 0$$

$$D_f(P \parallel Q) \geq 0.$$

When will the Divergence be 0

Suppose f is strictly convex at $u=1$

Equality in Jensen's Inequality occurs only when the RV $R(x)$ is constant Q -almost surely.

$$D_f(P \parallel Q) = 0$$

implies.

$$\frac{p(x)}{q(x)} = c \quad \mathbb{Q}\text{-almost Surely.}$$

we know that $\mathbb{E}_{\mathbb{Q}} \left[\frac{p(x)}{q(x)} \right] = 1$

$$\therefore c = 1$$

$$\frac{p(x)}{q(x)} = 1 \quad \mathbb{Q}\text{-almost Surely.}$$

which means $p(x) = q(x)$ almost everywhere

$$\therefore D_f(P \parallel Q) = 0 \iff P = Q.$$

Examples of f -Divergences:

$$D_f(P \parallel Q) = \int q(x) f\left(\frac{p(x)}{q(x)}\right) dx$$

1) choose $f(u) = u \log u$

$$\begin{aligned} D_f(P \parallel Q) &= \int q(x) \left[\frac{p(x)}{q(x)} \log\left(\frac{p(x)}{q(x)}\right) \right] dx \\ &= \int p(x) \log\left(\frac{p(x)}{q(x)}\right) dx \\ &= D_{KL}(P \parallel Q) \end{aligned}$$

2) Choose $f(u) = -\log u$

$$D_f(P \parallel Q) = \int q(x) \left[-\log \left(\frac{p(x)}{q(x)} \right) \right] dx$$

$$= \int q(x) \left[\log \left(\frac{q(x)}{p(x)} \right) \right] dx$$

$$= D_{KL}(Q \parallel P)$$

Reverse KL

3) Choose $f(u) = \frac{1}{2} |u-1|$

$$\begin{aligned} D_f(P||Q) &= \int q(x) \frac{1}{2} \left| \frac{p(x)}{q(x)} - 1 \right| dx \\ &= \int q(x) \frac{1}{2} \left| \frac{p(x) - q(x)}{q(x)} \right| dx \\ &= \int \frac{1}{2} |p(x) - q(x)| dx \\ &= D_{TV}(P, Q) \end{aligned}$$

Total variation distance.

4) Choose $f(u) = \frac{1}{2} u \log\left(\frac{2u}{u+1}\right) + \frac{1}{2} \log\left(\frac{2}{u+1}\right)$

$$\frac{1}{2} \frac{p(x)}{q(x)} \log \frac{2p(x)}{p(x)+q(x)} + \frac{1}{2} \log \frac{2q(x)}{p(x)+q(x)}$$

$$D_f(P \parallel Q) = \int q(x) f\left(\frac{p(x)}{q(x)}\right) dx$$

$$= \frac{1}{2} \int p(x) \log \frac{p(x)}{\left(\frac{p(x)+q(x)}{2}\right)} dx + \frac{1}{2} \int q(x) \log \frac{q(x)}{\left(\frac{p(x)+q(x)}{2}\right)} dx$$

$m(x)$ $m(x)$

$$M = \frac{P+Q}{2}$$

$$m(x) = \frac{p(x)+q(x)}{2}$$

$$= \frac{1}{2} \int p(x) \log \frac{p(x)}{m(x)} dx + \frac{1}{2} \int q(x) \log \frac{q(x)}{m(x)} dx$$

$$= \frac{1}{2} D_{KL}(P \parallel M) + \frac{1}{2} D_{KL}(Q \parallel M)$$

$$= D_{JSD}(P \parallel Q)$$

Why is D_{KL} not a Distance Metric

A distance Metric Should satisfy 4 properties.

Suppose the dist b/w 2 objects P & Q is $d(P, Q)$, then $d(\cdot, \cdot)$ is a distance metric if and only if it satisfies.

1) Non - Negativity.

$d(P, Q) \geq 0$, D_{KL} satisfies this

2) Identity of Indiscernibles

$d(P, Q) = 0 \iff P = Q$ D_{KL} satisfies this

3) Symmetry

$$d(P, Q) = d(Q, P)$$

KL divergences does not satisfy this $D_{KL}(P \parallel Q) \neq D_{KL}(Q \parallel P)$

Consider

$$p = (0.9, 0.1)$$

$$q = (0.5, 0.5)$$

$$\left. \begin{array}{l} p = (0.9, 0.1) \\ q = (0.5, 0.5) \end{array} \right\} \begin{array}{l} D_{KL}(P \parallel Q) = 0.9 \log \frac{0.9}{0.5} + 0.1 \log \frac{0.1}{0.5} \approx 0.368 \\ D_{KL}(Q \parallel P) \approx 0.511 \end{array}$$

$$D_{KL}(P \parallel Q) \neq D_{KL}(Q \parallel P)$$

$$D_{KL}(P||Q) = \mathbb{E}_P \left[\log \frac{P}{Q} \right]$$

Regions where P is large matter a lot, Regions where P is small hardly matters.

$D_{KL}(P||Q)$: How Surprised is the model about Samples that Actually Occurs.

$D_{KL}(Q||P)$: How much prob does the model assigns to regions.
where data rarely occurs.

4) Triangular Inequality

P

Q R

$$d(P, R) \leq d(P, Q) + d(Q, R)$$

going directly from P to R can never be longer than going through Q.

$$p = (0.1, 0.9)$$

$$q = (0.5, 0.5)$$

$$r = (0.9, 0.1)$$

$$D_{KL}(P, R) \approx 1.758$$

$$D_{KL}(P, Q) \approx 0.368$$

$$D_{KL}(Q, R) \approx 0.511$$

$$\left. \begin{array}{l} 0.368 \\ 0.511 \end{array} \right\} 0.879$$

$$1.758 \geq 0.879 \quad \blacksquare$$

is Total variation Distance a valid Dist metric?